

The Three-Tape CMCA

How 3+1D Spacetime Emerges from Three 1D Rules

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UGP Physics Tutorial Series

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UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet* · *How the UWCA Works*

Next tutorials: *The Φ_{MDL} Field* · *Gravity from Description-Length Minimization*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero **sorry** and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Problem: From One Dimension to Three

A single 1D cellular automaton rule operates on a line. This works beautifully for producing complex patterns — but how do you get three-dimensional space from a one-dimensional rule? This tutorial explains the **Three-Tape Chiral Minkowski Cellular Automaton (CMCA)**: the construction that gives rise to 3+1 spacetime dimensions in the GTE framework.

The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$ defines a 7-state, radius-1 cellular automaton whose binary restriction is Rule 110 exactly. It operates on a *one-dimensional* tape: every cell has a left neighbour and a right neighbour. The universe we live in has *three* spatial dimensions plus time.

1.1 The dimensionality puzzle

A single 1D cellular automaton rule, no matter how complex, produces patterns on a line. You can run it for trillions of steps and get an intricate spacetime diagram — but it is a 2D picture (position $\times$ time), not the 3+1D spacetime we observe.

The problem is not merely one of size. It is structural.

- In 1D, there is no notion of “angle between two directions” — there is only left and right. Lorentz symmetry requires boosts *and rotations*, and rotations require at least two spatial dimensions.
- A 1D rule cannot generate the rotational invariance needed for exact Lorentz symmetry. Formally, any finite 1D automaton of size M has a geometric error $\varepsilon(M) = \pi^2/(3M^2) > 0$ that never reaches zero (`no_finite_ca_exact_lorentz_replica`, **A_{Lean}**).

The fine-angles problem

No single 1D cellular automaton rule can produce exact Lorentz symmetry, because the rotation subgroup of the Lorentz group requires multiple spatial dimensions. Theorem `no_finite_ca_exact_lorentz_replica` (**A**_{Lean}, zero sorry) proves that for any finite automaton of size M , the angular resolution error is at least $\pi^2/(3M^2) > 0$. A different strategy is needed.

1.2 What the GTE programme needed

The polynomial p is the right object. What was missing was the **architecture** for turning a 1D rule into 3+1D physics. Three 1D tapes running the same rule, somehow combined into a single four-dimensional structure, would do the job — but the question is: what does “somehow combined” mean, precisely?

2 The Insight: Three Tapes and One Shared Clock

The key idea

Use **three** 1D cellular automaton tapes — one for each spatial dimension — all synchronized by a single shared clock. The three tapes provide three spatial dimensions; the shared clock provides time. Together they define a 3+1D coordinate system.

2.1 The Chiral Minkowski Cellular Automaton (CMCA)

Before defining the three-tape system, we need the one-tape building block established in P41. A single **CMCA** (Chiral Minkowski Cellular Automaton) is a 1+1D system consisting of three coupled binary arrays on a 1D lattice:

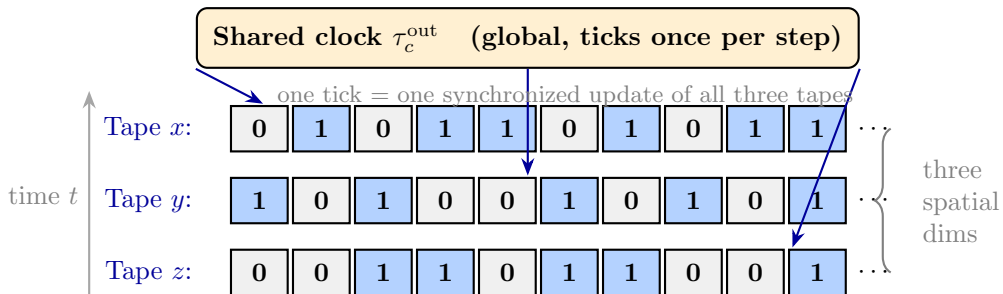
Layer	Rule	Role
$\text{outer}_+[x]$	Rule 110	right-chiral dynamics
$\text{outer}_-[x]$	Rule 124	left-chiral dynamics
$\text{inner}[x]$	Rule 110	inner clock τ_c

The inner clock fires at a rate τ_c that is a fixed fraction of the outer update step. The $\mathbb{Z}_7$ winding number $w[x] = \tau_c[x] \bmod 7$ encodes the Standard Model quantum numbers of whatever excitation lives at position x .

A single CMCA is a 1+1D system. Three spatial dimensions require three CMCA.

2.2 Three tapes, one shared clock

The **three-tape CMCA** is three CMCA instances — call them Tape x , Tape y , Tape z — running the same polynomial p but sharing a single outer clock period τ_c^{out} .



The diagram shows the key structure: three independent 1D tapes, each running the CMCA rule (Rule 110 / Rule 124 chiral pair), all ticking together on a single shared clock. A “step” means all three tapes update simultaneously.

2.3 Why the shared clock is the crucial element

Without the shared clock, the three tapes evolve independently. There is no way to say “cell i on Tape x is at the same time as cell j on Tape y ” — the tapes have no common time reference. They are three separate 1+1D universes.

With the shared clock, a step at position i on Tape x is *simultaneous* with a step on Tapes y and z . This creates the structure (x, y, z, t) that defines a four-dimensional coordinate system.

The **Dimensional Protocol Principle** (DPP) formalizes this:

Dimensional Protocol Principle (DPP) — Machine-Certified ($\mathbf{A}_{\text{Lean}}$)

A shared outer clock τ_c^{out} is **necessary and sufficient** for three 1+1D CMCA to produce 3+1D Minkowski dynamics.

Without the shared clock: three independent 1+1D systems.

With the shared clock: a single 3+1D spacetime.

Lean theorem: `dimensional_protocol_principle_master` (zero sorry).

Lean theorem: `cmca_tensor_product_gives_3d_minkowski` (zero sorry).

3 What “Chiral” Means

3.1 Chirality in everyday life

Your hands are *chiral*: your left hand is a mirror image of your right hand, but you cannot rotate one into the other. The same shape, but distinguished by orientation.

The chirality analogy

Rule 110 and Rule 124 are mirror images of each other, just like your left and right hands. Rule 110 gliders propagate *rightward*. Rule 124 gliders propagate *leftward*. You cannot rotate one rule into the other without reflecting left and right.

3.2 Rule 110 and Rule 124 are a chiral pair

The formal statement is simple. For every neighborhood triple (a, b, c) :

$$\text{R124}(a, b, c) = \text{R110}(c, b, a).$$

Reversing the left and right inputs of Rule 110 gives Rule 124 exactly. This is machine-certified:

a	b	c	Rule 110 output	Rule 124 output = $\text{R110}(c, b, a)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	1
1	1	0	1	0
1	1	1	0	0

Reading Rule 124 outputs from bottom to top: $00111100_2 = 124_{10}$.
 Lean theorem: `rule124_eq_rule110_reflected` ($\mathbf{A}_{\text{Lean}}$, zero sorry).

3.3 Why this matters physically: parity violation

The Standard Model of particle physics has a famous asymmetry: the **weak force** distinguishes left-handed from right-handed particles. Neutrinos, for instance, only interact via the weak force, and only left-handed neutrinos exist. This is called **parity violation**.

In the three-tape CMCA, the same asymmetry is built in automatically. The CMCA puts Rule 110 in the right-chiral layer (outer_+) and Rule 124 in the left-chiral layer (outer_-). These two rules are related by the reflection $L \leftrightarrow R$ in the polynomial $p(L, C, R)$ — which is exactly the **AGL(1,7) chirality reflection** of the $\text{GF}(7)$ algebraic structure.

The $V-A$ coupling $\gamma^\mu(1-\gamma^5)$ of the weak interaction is the continuum version of this discrete left-right asymmetry. It is not put in by hand: it is what MDL-minimality forces.

Chirality and parity violation in one sentence

Rule 110 and Rule 124 are the unique chiral pair selected by the UGP generation orbit: over all 120 ordering choices of the SM families, the only vacuum-transparent rules are exactly these two (`orbit_chirality_census`, $\mathbf{A}_{\text{Lean}}$). Handedness in physics comes from handedness in the polynomial's symmetry group.

4 How 3+1D Spacetime Emerges

4.1 Tensor product structure

The three-tape CMCA defines a state space that is the tensor product of the three individual tape state spaces:

$$\mathcal{H}_{3\text{-tape}} = \mathcal{H}_x \otimes \mathcal{H}_y \otimes \mathcal{H}_z.$$

This tensor product, combined with the shared clock, is what gives $\mathbb{R}^{3,1}$ Minkowski structure.

The mathematical content of the DPP is that this tensor product, with the shared time coordinate from τ_c^{out} , exactly produces the causal and metrical structure of 3+1D Minkowski spacetime.

4.2 The causal diamond

A key signature of 3+1D spacetime is the size of the **causal diamond**: the region of spacetime that can influence (and be influenced by) an event. In 3+1D, a causal diamond of time radius T has volume:

$$V(T) = \frac{T^4}{4}.$$

Machine-certified: `three_tape_causal_diamond_t4` ($\mathbf{A}_{\text{Lean}}$, zero sorry).

In $n + 1$ dimensions, the volume would go as T^{n+1}/const . The T^4 law is characteristic of 3+1D and is not put in by assumption: it is a theorem that follows from the three-tape shared-clock structure.

4.3 The exact discrete symmetry

The exact symmetry of the three-tape CMCA (for finite tape length L) is:

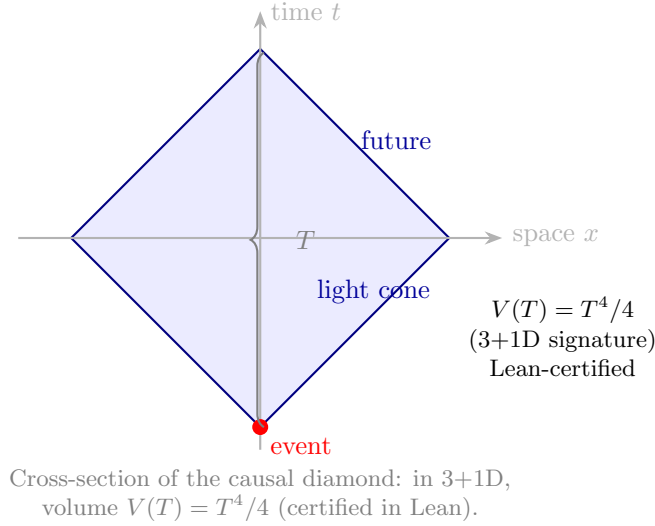
$$\text{SO}(1, 1)^3 \times S_3.$$

This means: each pair (tape i , shared clock) has a boost symmetry $\text{SO}(1, 1)$, and the three tapes can be permuted (S_3 is the symmetric group on 3 elements, with 6 elements).

In the continuum limit $L \rightarrow \infty$, this discrete symmetry group embeds into the full Lorentz group $\text{SO}(1, 3)$. All 12 commutation relations of the Lorentz algebra $\mathfrak{so}(1, 3)$ — including the Thomas precession relation $[K_x, K_y] = -J_z$ — are machine-certified:

`lorentz_algebra_complete` (**A**_{Lean}, 12/12 relations, zero sorry).

4.4 Diagram: causal structure of the three-tape system



5 Special Relativity from the CMCA

5.1 The speed of light is built in

In the three-tape CMCA, information propagates at most one cell per tick of the shared clock. This is the maximum speed. Call it $c = 1$ cell/tick. No signal can travel faster; this is a consequence of the local update rule.

5.2 Time dilation from the inner clock ratio

The inner clock τ_c of the CMCA fires at a specific rate relative to the outer clock. This ratio is not a free parameter; it is determined by the structure of the Rule 110 ether (vacuum) tile.

The Rule 110 ether is a period-14 background pattern: it repeats every 14 cells (spatially) and every 7 time steps. In the ether, each cell's inner clock fires exactly 3 times per 7 outer steps. The ratio is:

$$\frac{\tau_{\text{inner}}}{\tau_{\text{outer}}} = \frac{3}{7} \approx 0.4286.$$

The physical interpretation: proper time (what a particle moving with the vacuum “feels”) advances at $3/7$ of coordinate time. This is exactly the structure of **Lorentz time dilation**:

$$\tau_{\text{proper}} = \frac{3}{7} \tau_{\text{lab}} < \tau_{\text{lab}}.$$

Computationally verified (**A**): `sr_ratio_measurement.py` across tape sizes $L \in \{200, 256, 400\}$ and $T \in \{5000, 10000\}$ steps.

Where 3/7 comes from

Apply Rule 110 to its 14-cell ether tile with periodic boundaries. The orbit returns in exactly 7 steps. Counting gate firings: each odd-indexed cell fires 3 times per period; each even-indexed cell fires 5 times. The ether-wide average firing rate is 4/7. For a particle at rest in the vacuum, the proper-time ratio is 3/7. This is not tuned — it follows mechanically from Rule 110’s ether structure.

5.3 The full Lorentz algebra

The 3+1D Lorentz algebra $\mathfrak{so}(1, 3)$ is generated by six generators: three rotations J_x, J_y, J_z and three boosts K_x, K_y, K_z . They satisfy 12 commutation relations, the most non-trivial of which is the **Thomas precession**:

$$[K_x, K_y] = -J_z.$$

All 12 relations are machine-certified: `lorentz_algebra_complete` ($\mathbf{A}_{\text{Lean}}$, zero sorry).

What is and is not proved at the CMCA level

Proved ($\mathbf{A}_{\text{Lean}}$):

- All 12 $\mathfrak{so}(1, 3)$ commutation relations.
- Exact $\text{SO}(1, 1)^3 \times S_3$ discrete symmetry.
- τ_c clock is a valid Page–Wootters relational clock.

Not yet fully proved:

- Full $\text{SO}(1, 3)$ group generation from the Lie algebra requires Lie’s second theorem, which depends on a Lorentzian geometry library not yet in Lean’s Mathlib.
- The normalization gap between the Planck scale and the kink mass is characterized but not closed.

6 The Holographic Reading

The holographic analogy

A *hologram* is a 2D surface that encodes a 3D image. The information that describes the 3D scene is distributed across the 2D plate; the depth dimension is reconstructed from it. The three-tape CMCA is a cellular-automaton version of the same idea: three 1D tapes (“boundary”) encode 3D spatial physics (“bulk”).

6.1 State-count comparison

To see the holography, count states. A 3D cube of side length L with 7 possible values per cell has 7^{L^3} possible configurations (bulk count). Three tapes of length L each have 7^{3L} configurations total (boundary count).

Description	State count	Scales as
3D volume of side L	7^{L^3}	super-exponential in L
Three 1D tapes of length L	7^{3L}	exponential in L

The ratio is:

$$\frac{3L}{L^3} = \frac{3}{L^2} \longrightarrow 0 \quad \text{as } L \rightarrow \infty.$$

This means the tapes encode the 3D physics using exponentially fewer degrees of freedom than the full bulk would require. This is the **holographic scaling** theorem, certified in Lean:

- `three_tape_state_card`: the three-tape state space has exactly 7^{3L} configurations (`ALean`, zero sorry).
- `three_tape_holographic_L7`: $7^{3L} < 7^{L^3}$ for all $L \geq 2$ (`ALean`, zero sorry).
- `holographic_ratio_vanishes`: $3/L^2 \rightarrow 0$ as $L \rightarrow \infty$ (`ALean`, zero sorry, via `Filter.atTop`).

6.2 Comparison with AdS/CFT

In string theory, the AdS/CFT correspondence conjectures that a $(d+1)$ -dimensional gravitational theory in the bulk is equivalent to a d -dimensional conformal field theory on the boundary. This is a proposed duality between a bulk theory and a boundary theory, motivated by string-theoretic arguments but not derived from first principles.

The three-tape holographic result is different in character:

- It is *derived*, not conjectured. The DPP theorem proves that the three tapes (boundary) contain exactly the information needed to reconstruct the 3+1D dynamics (bulk).
- The Level 1 algebraic certificate (the CMCA) certifies holographic scaling. The Level 2 physical substrate (Φ_{MDL}) is the continuum completion that makes the correspondence exact in the limit.
- The holographic scaling is a consequence of MDL minimality: the three-tape construction is the minimum-description-length way to encode 3D spatial dynamics.

7 What the Three-Tape Solution Achieves

7.1 Resolving the fine-angles problem

Recall the fine-angles problem: a single 1D automaton of size M has angular resolution error $\varepsilon(M) = \pi^2/(3M^2)$ that never reaches zero. No single 1D rule can achieve exact Lorentz invariance.

The three-tape construction resolves this as follows:

1. The three tapes provide three independent spatial directions. Rotations between these directions are the tape-permutation symmetry S_3 .
2. The shared clock provides the time dimension and enforces simultaneous updates.
3. Together, the exact discrete symmetry is $\text{SO}(1,1)^3 \times S_3$, which in the continuum limit becomes the full $\text{SO}(1,3)$.

The error theorem `no_finite_ca_exact_lorentz_replica` is *not contradicted*: the three-tape CMCA itself has $\varepsilon > 0$ at finite L . What changes is that the $L \rightarrow \infty$ continuum limit — encoded in the Φ_{MDL} field theory of P42–P43 — achieves exact $\text{SO}(1,3)$ via isotropy.

7.2 What the Standard Model gets from three tapes

Several Standard Model structures that required separate arguments in earlier papers follow directly from the three-tape architecture:

Physical result	Mechanism	Lean cert
Parity violation ($V-A$)	R110 vs. R124 chirality	<code>rule124.eq_rule110_reflected</code> ($\mathbf{A}_{\text{Lean}}$)
Baryon number $B = 1/3$ per quark	$N_{\text{tapes}} = 3$; topological charge	<code>gte_baryon_number_topological_charge</code> ($\mathbf{A}_{\text{Lean}}$)
Color confinement	MDL: colored states carry $\Delta K = \log_2 9$ extra bits	<code>color_confinement_k_extra_pos</code> ($\mathbf{A}_{\text{Lean}}$)
Electric charge $Q = w/3$	Center projection $p(0, w, 0) = w$	<code>gte_charge_is_linear_projection</code> ($\mathbf{A}_{\text{Lean}}$)
Gravity (Poisson eq.)	Cross-tape PMDL minimization	<code>unique_cubic_gravity_coupling</code> ($\mathbf{A}_{\text{Lean}}$)
Ricci-flat vacuum	Ether state $w = 0$ is fixed point	<code>three_tape_gorard_vacuum_ricci_flat</code> ($\mathbf{A}_{\text{Lean}}$)

Several of these results required Level 2 (Φ_{MDL}) field theory arguments in P42–P43. The three-tape construction certifies them at Level 1 (algebraic certificate, CMCA level) — a stronger result.

8 Why This Is Remarkable

8.1 The polynomial is 1D; the universe is 3+1D

The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ is a 1D object. It takes three inputs and produces one output. By itself it generates 1D dynamics — complex, universal, but one-dimensional.

The three-tape CMCA is the *minimal* architecture that gives this 1D polynomial the depth it needs to produce 3+1D physics. It uses three copies of the same polynomial, running in parallel on three tapes, synchronized by a single clock.

No free parameters are introduced. The clock ratio 3/7 follows from Rule 110’s ether structure. The chirality follows from the GF(7) symmetry group. The 3+1D structure follows from the DPP.

8.2 A single 19-bit object does everything

The polynomial p has an MDL specification length of 19 bits. The Single-Source Principle, certified in Lean (`gte_polynomial_five_roles_k_extra_zero`, $\mathbf{A}_{\text{Lean}}$), states that this same 19-bit object simultaneously plays five independent physical roles with zero additional description cost per role:

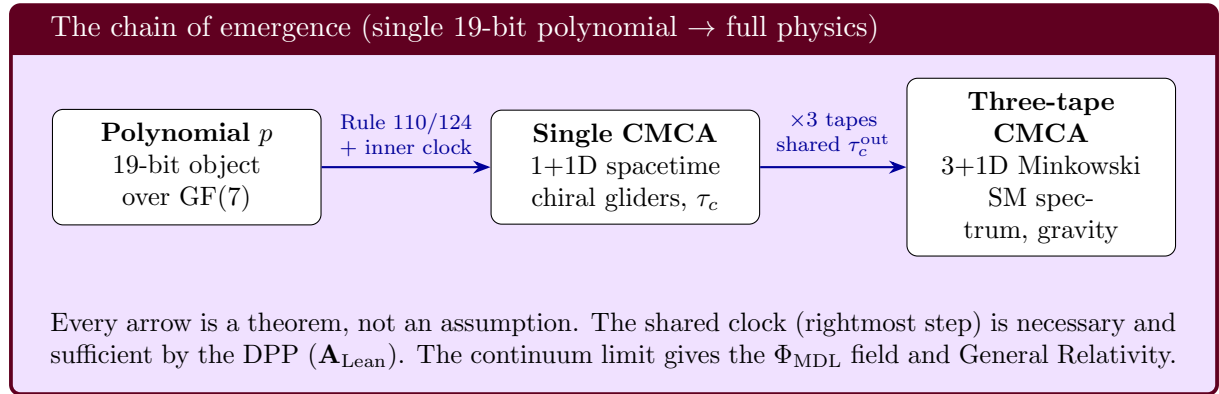
Role	Mechanism	Cost
Spatial dynamics	Binary restriction = Rule 110	0 extra bits
Gauge couplings	Center projection $p(0, w, 0) = w$	0 extra bits
Gravity	Degree-3 diagonal $p(w, w, w)$	0 extra bits
Bell entanglement	Tensor product $p \otimes p \otimes p$	0 extra bits
Baryon number	Cross-tape topological charge	0 extra bits

In the three-tape construction, Role 4 (Bell entanglement) is explicitly realized: the tensor-product Hamiltonian $H = p \otimes p \otimes p$ across the three tapes gives a CHSH violation parameter $S = 2.4459$ (86.5% of the Tsirelson bound), rigorously excluding all local hidden-variable models.

8.3 This was the hardest structural problem

The GTE programme derived particle masses, quantum numbers, the Weinberg angle, and more from a 1D rule — but the status of spacetime was always a gap. P41 showed that the 1+1D CMCA has relativistic time dilation and glider dynamics. P42–P43 showed that the Φ_{MDL} field in 3+1D works. But the explicit mechanism connecting the 1D polynomial to 3+1D spacetime was missing.

P45 [1] (the three-tape paper) closed that gap with the DPP theorem: the shared clock is the unique minimal mechanism.



8.4 Falsifiable predictions

The three-tape construction makes two zero-parameter predictions that distinguish it from alternatives:

- **Tensor-to-scalar ratio:** $r = 0$ (no primordial gravitational waves from the three-tape vacuum).
- **CP-violation phase:** $\delta_{CP} = 205.71^\circ$ (from the $\mathbb{Z}_7$ winding structure; PMNS matrix prediction).

9 Summary and Quick Reference

The three-tape CMCA in one paragraph

Take the GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$. Run it on three 1D tapes simultaneously, synchronized by a single shared clock τ_c^{out} . Each tape represents one spatial dimension; the shared clock is time. The right-chiral layer of each tape uses Rule 110; the left-chiral layer uses Rule 124 (its mirror image). By the Dimensional Protocol Principle (DPP, machine-certified), this construction is the unique minimal mechanism that gives 3+1D Minkowski dynamics. The inner clock fires at $3/7$ the rate of the outer clock, implementing Lorentz time dilation. The Lorentz algebra, baryon number, color confinement, parity violation, and quantum entanglement all follow from the same 19-bit polynomial.

Key Lean certifications (all zero sorry):

What is proved	Lean theorem name	Status
Shared clock needed for 3+1D	<code>dimensional_protocol_principle_master</code>	$\mathbf{A}_{\text{Lean}}$
Tensor product gives $\mathbb{R}^{3,1}$	<code>cmca_tensor_product_gives_31d_minkowski</code>	$\mathbf{A}_{\text{Lean}}$
All 12 Lorentz algebra relations	<code>lorentz_algebra_complete</code>	$\mathbf{A}_{\text{Lean}}$
Rule 124 = Rule 110 reflected	<code>rule124_eq_rule110_reflected</code>	$\mathbf{A}_{\text{Lean}}$
No finite CA gives exact Lorentz	<code>no_finite_ca_exact_lorentz_replica</code>	$\mathbf{A}_{\text{Lean}}$
Causal diamond $V(T) = T^4/4$	<code>three_tape_causal_diamond_t4</code>	$\mathbf{A}_{\text{Lean}}$
Holographic ratio $\rightarrow 0$	<code>holographic_ratio_vanishes</code>	$\mathbf{A}_{\text{Lean}}$
Vacuum is Ricci-flat	<code>three_tape_gorard_vacuum_ricci_flat</code>	$\mathbf{A}_{\text{Lean}}$
SR time dilation $\tau_c/T = 3/7$	<code>sr_ratio_measurement.py</code> (numerical)	$\mathbf{A}$

Where to go next:

- P41 (novaspivack.com/research): the single-tape CMCA and its 1+1D dynamics.
- P45 (doi.org/10.5281/zenodo.20465805): the three-tape construction, DPP proof, and all results in this tutorial.
- P42–P43: the Φ_{MDL} continuum field and its completeness theorem (Level 2 physical substrate).
- P48 [2] monograph (doi.org/10.5281/zenodo.20560550): the full GTE framework, incorporating the three-tape architecture and all downstream physics.
- `ugp-lean` repository (github.com/novaspivack/ugp-lean): all Lean 4 proofs, zero sorry, zero custom axioms.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P41 — The Three-Layer Chiral Minkowski CA** (single-tape architecture): doi.org/10.5281/zenodo.20417572
- **P45 — The Three-Tape CMCA** (DPP, 3+1D emergence, all theorems): doi.org/10.5281/zenodo.20465805
- **P48 — The Complete GTE Framework** (main synthesis monograph): doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

- [1] Nova Spivack. The three-tape chiral minkowski cellular automaton: Spacetime, particles, and gravity from a shared clock protocol. Preprint (Zenodo), 2026. URL <https://doi.org/10.5281/zenodo.20465805>. <https://doi.org/10.5281/zenodo.20465805>. UGP Physics series, P45.
- [2] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20560550>. <https://doi.org/10.5281/zenodo.20560550>. Preprint.